

5. (a) Metal ores are the materials found in the Earth's crust from which metals can be extracted. The following table gives information about some common metal ores.

Name of metal ore	Metal extracted from ore	Formula of ore
anglesite	lead	PbCl_2
copper pyrite	copper	CuFeS_2
cryolite	aluminium	Na_3AlF_6
salt petre	potassium	KNO_3
syberite	gold	AgAuTe_2
tin pyrite	tin	$\text{Cu}_2\text{FeSnS}_4$

Use the information in the table to give

- (i) the **total** number of atoms shown in the formula of cryolite, [1]

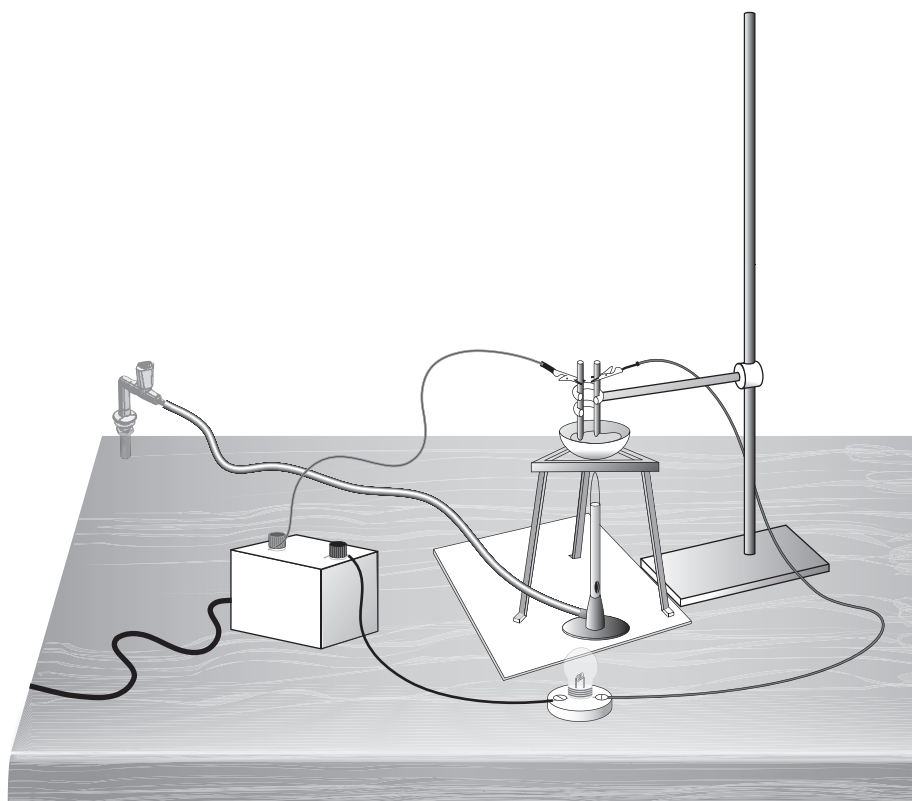
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- (ii) the name of the ore that contains a ratio of 1:4 metal to non-metal atoms. [1]

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- (b) The following diagram shows how the electrolysis of anglesite (lead chloride) can be carried out in the laboratory.



- (i) Give the reason why lead chloride needs to be melted for electrolysis to take place. [1]

- (ii) Explain why the chloride ions move towards the positive electrode. [2]

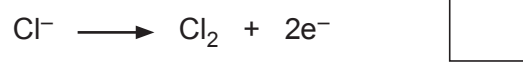
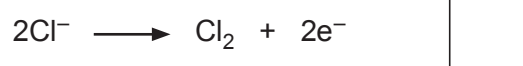
- (iii) The process occurring at the negative electrode is shown by the following equation.



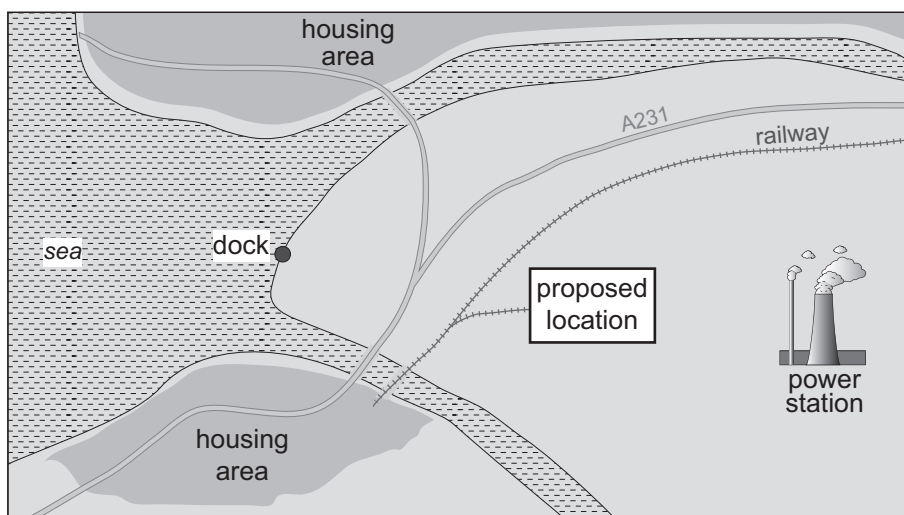
Use this **equation** to state what is meant by reduction. [1]



- (iv) Tick (✓) the equation that shows the reaction taking place at the positive electrode. [1]



(c) The following diagram shows a proposed location for a new aluminium works in Wales.



Use the information in the diagram to explain why this would make a suitable location for an aluminium works. [6 QER]

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